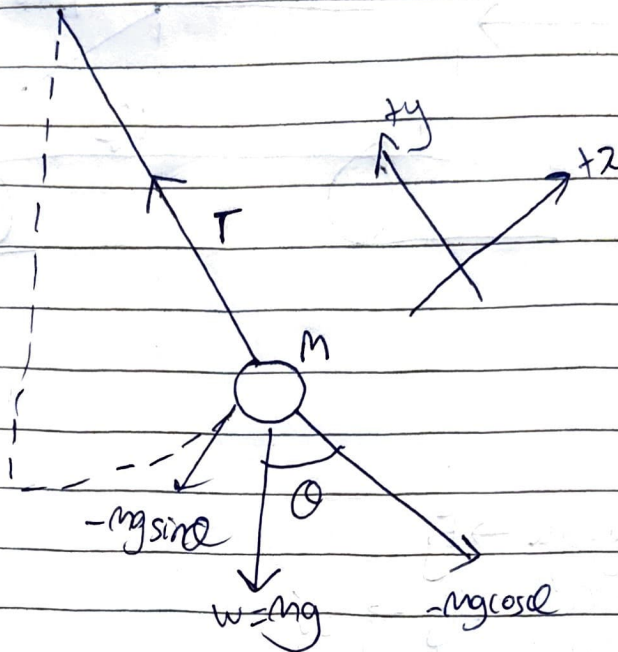


H-066
H-067
H-090
H-094

Free Body - Diagram

FBD 1: LOAD



weight (downward)

$$W = Mg$$

Tension (Along rope)

T

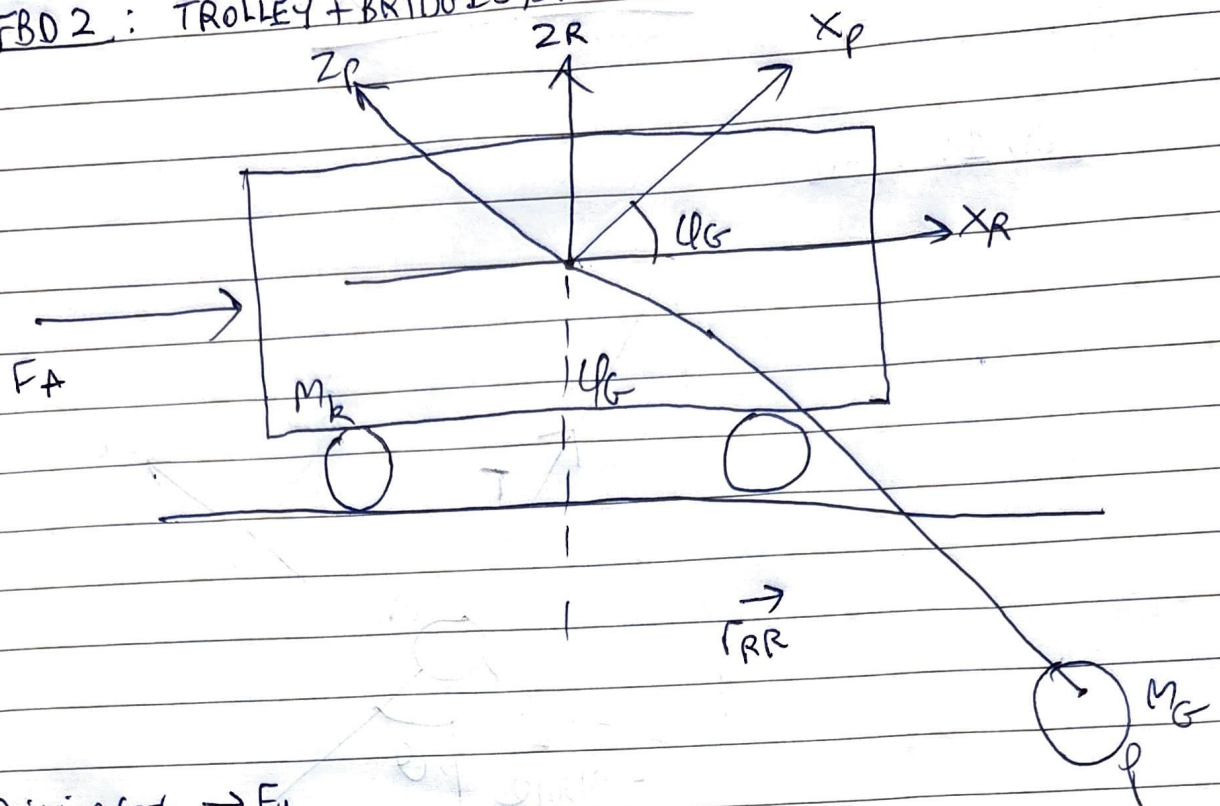
Angle $\rightarrow \theta$

$T \cos \theta$ (vertical)

$T \sin \theta$ (horizontal)

Horizontal acceleration = y

FBD 2: TROLLEY + BRIDGE SYSTEM



- 1) Driving force $\rightarrow F_y$
- 2) Friction/Damping $\rightarrow c_y \dot{y}$
- 3) Inertial force from load

FBD 3:

